**EXIM Bank Agricultural University Bangladesh (EBAUB), Chapa Nawabganj**

**Department of Computer Science & Engineering (CSE)**

**Semester**: 2nd **Batch**: 3rd

**Course Tittle:** Basic Programming Sessional (CSE 0611 1212)

## Programming Part General:

1. Write a C program to print **the ASCII value**.
2. Write a C program to print the **ASCII value of any number**.
3. Write a C program to print **"Hello World" without using a semicolon**.
4. Write a C program to **convert Fahrenheit to Celsius**.
5. Write a C program to **convert feet to meter**. ***(Hint:** 1 ft = 3.28 m)*
6. Write a C program to **find the square root of a number**.
7. Write a C program to **calculate the area of a circle.**
8. Write a C program to calculate the **area of a rectangle**. **Formula:** Area = length × width
9. Write a C program to calculate the **volume of a cube**. **Formula:** Volume of cube = $a^3$
10. Write a C program to calculate the **surface area and volume of a cone**. **Formula:** Surface area = $\pi r ( r + \sqrt{(r^2 + h^2)} )$ & Volume = $(1/3) \times \pi \times r^2 \times h$
11. Write a C program to **swap two numbers using a temporary variable**.
12. Write a C program to **swap two numbers without using a temporary variable**.
13. Write a C program to **swap two numbers** using **pointers**.
14. Write a C program to **find the area of a triangle**.
15. Write a C program to **find the area of a triangle** using **Heron's formula**.
16. Write a C program to **find the last two digits of a given number**.
17. Write a C program to **calculate the interior angle of a regular polygon**. **Formula:** Angle = [ (n - 2) × 180° ] / n
18. Write a C program that **takes an integer input (0–9) and prints it in words** using the **switch** statement.
19. Write a C program to **add two numbers without using the addition operator (+)**.
20. Write a C program to **subtract two numbers without using the subtraction operator (−)**.
21. Write a C program to **find the largest number among three numbers** using **conditional statements**.

## Conditional Programming:

1. Write a program to **check odd or even number** using the **conditional (ternary) operator**.
2. **Convert kilometer (km) into meter, centimeter, or millimeter**.
3. C program to **find the maximum of two numbers**.
4. C program to **check a given number is positive or negative**.
5. C program to **check whether a triangle is valid or not**, if the three sides are given.
6. C program to **accept a number from the user and print if it is multiple of 7**.
7. C program to **check whether a given year is a leap year or not**.
8. Write a C program to **calculate the roots of a quadratic equation**.
9. Write a C program to **perform division of two numbers**.
10. Write a C program to **sum two numbers**, **find the quotient**, and print a message if the **average is less than the sum**.

11. **Write a C program to check the relationship between two numbers** *(greater than, less than, or equal to).*
12. Write a C program for a university admission system that **accepts students who passed Mathematics and English and either Chemistry or Physics**.
13. Write a C program to **add all the numbers entered by the user until the user enters 0**.
14. Write a C program to **print a series from 20 to 1 using nested loops**.
15. Write a C program to **print a descending numerical pattern using nested loops**.

## Looping Program:

1. Write a program to **print numbers from 1 to N**.
2. Write a C program to **find the factors of a given number n**.
   **Example:** Input = 10, Output = factors are 5 and 2.
3. Write a program to **find the summation of the series**:
   **1 + 2 + 3 + 4 + … + n**.
4. Using C, write a program to **find the sum of the following series**:
   **1 + 1/2 + 1/3 + 1/4 + 1/5 + … + 1/n**
5. Write a C program to **find the maximum price from 20 items**.
6. **How many bits have to change to convert integer A to integer B**.
   **Sample:** A = 10, B = 20, Output = 4
7. Write a program to **find the sum of odd numbers from 1 to n**
   *Or* **1 + 3 + 5 + 7 + … + n**
8. Write a program for **1 + 2 + 3 + … + n** with **O(1) time complexity and it will take n as input.**
9. Write a C program to **find the sum of even numbers between 1 and n**
   *Or* **2 + 4 + 6 + 8 + … + n**
10. Write a program to **find the value of the series**:
    **(1×1) + (2×2) + (3×3) + (4×4) + (5×5) + … + (n×n)** *Or* **$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + … + n^2$**
11. Write a program to **calculate the value of the series**:
    **$1^2 – 3^2 + 5^2$ - … + $(n^2)$**
12. Write a C program to **calculate the factorial of a number using a while loop**.
13. Write a program to **check a given number is an *Armstrong number* or not.**
14. Write a C program to **check a given number is a *Perfect number* or not.**
15. C program to **add the digits of a number**.
16. How **calculate the power ($x^y$) of a number in C**.
17. Write a C program to **check a given number is a *Palindrome number* or not**.
18. Write a C program to **check whether a given number is a *Strong number* or not**.
19. Write a C program to **reverse a given number**.
20. Write a C program to **convert binary to decimal**.
21. C program to **count the total number of digits of a number using for loop**.
22. Write a C program to **check a given number is prime or not**.
23. Write a C program to **print all prime numbers between 1 and N**.
24. A prime number is a number that is evenly divided by only 1 and itself. Write a program to your favorite language to **print the first 100 prime numbers**.
25. Write a program to **generate the nth Fibonacci series in C**.
26. Write a C program to **print Floyd's Triangle**.
27. Write a C program to **print the following output sequence**:
    **1 2 3 6 9 18 27 54 81 162**

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28. Write a C program to **take marks of students and print the average marks**.
29. Write a C program to **print the integers from 1 to 6 and their cubes**.
30. Write a C program that **reads five numbers from the keyboard and displays the smallest number, the largest number, and the average of those numbers**.
31. Write a C program to **print the following number pattern**:
    1
    12
    123
    1234
    12345
32. Write a C program to **print the following pattern**:
    54321
    5432
    543
    54
    5
33. Write a C program to **print the following pattern**:
    12345
    1234
    123
    12
    1
34. Write a C program to **print the following star pattern**:
    *
    **
    ***
    ****
    *****
35. Write a C program to **print the following star pattern**:
    *
    ***
    *****
    *******
    *********
36. Write a program to **print a diamond pattern** as shown.
      *
     ***
    *****
     ***
      *
37. Write a C program to **find the GCD (HCF) and LCM of two numbers**.
38. Write a program that **reads n numbers from the keyboard and displays the maximum and minimum of those numbers**.
39. Write a C program to **print the upper triangular matrix**.
40. What will be the **output of the following program**?
    *(Hint: In case of address, write either the address of x or the address of y.)*
41. Write a program in C to **calculate the sum of the series**:
    **1 + (1 + 2) + (1 + 2 + 3) + … + (1 + 2 + … + n)**
42. Write a program to **evaluate the series**:
    **$1\times3 + 2\times5 + 3\times7 + \ldots + n\times(2n + 1)$**
43. Write a program to **calculate the value of the following series**:
    **$1 + 1/2^2 + 1/3^2 + 1/4^2 + \ldots + 1/n^2$**
44. Using the **C language**, write a program to **find the value of the series**:
    **$Y = 3 \times 9 \times 27 \times 81 \times \ldots \times 59049$**

## Array:

1. C program to **print all the elements of an array**.
2. Write a C program to **find the largest element in an array**.
3. Write down a **function to compute the sum of the row an n×m matrix of integer**.
4. Write a C program to **find the largest and smallest elements in an array**.

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5. Write a **Bubble Sort C program** that **sorts an array of integers**. Write down the **code that tests the sorting algorithm** written.
6. Write a C program to **sort a set of strings in alphabetical order**.
7. Write a program using **C** that **reads five numbers from the keyboard** and displays the **smaller number, larger number, and average** of these number.
8. Write a C program for **swapping two arrays.**
9. Write a program to **find the second maximum element in an array without using any library function**.
10. Write a C program to **multiply two matrices using multidimensional arrays.**
11. Write a C program to find **the sum of two matrices** using an **array function.**
12. C program to **increment every element of a given array by 1 and print the updated array.**

**Program using Recursive Function:**

1. Write a C program to **count the digits of a given number using recursion**.
2. Write a program in C to **find the sum of digits of a number using recursion**.
3. Write a C program to **reverse a number using recursion.**
4. Write a C program to **calculate the factorial of a number using recursion**.
5. Write a C code to **print the Fibonacci series using recursion**.
6. Write a C program to **check whether a number is prime or not using recursion**.
7. Write a program to **find the power of a number using a C program**.
8. Write a C program to **find the sum of digits using recursion**.
9. Write a program to **find the GCD of two numbers using recursion**.
10. Write a function **sumdigits** to **find the sum of digits of a number**.

**String:**

1. Write a program in C to **print a string**.
2. Write a C program to **accept a string from the user and print it**.
3. Write a C program to **accept a string and find the total number of characters**.
4. Write a C program to **convert lowercase letters to uppercase using a library function**.
5. Write a C program to **convert lowercase letters to uppercase without using a library function**.
6. Write a program to **find the length of a string**.
7. Write a C program to **reverse a string**.
8. Write a program to **count word in a string or a sentence without using library functions**.
9. Write a program to **copy one string value into another without using library functions**.
10. Write a program to **count all vowels present in the string.**
11. **Swapping of strings** using C program.
12. Write a C program to **check whether a given string is a palindrome or not**.
13. Write a program to **remove spaces from a sentence and convert the words into Camel Case**.

**File:**

1. Write a program to **print numbers from 1 to N using file handling**.
2. Take two numbers as input from **a file and store their sum in an output file**.
3. Take a number as input from **a file and calculate its factorial.**
4. A file named **myfile.txt** contains a list of days and the power shortage of each day like this:
   **1, 4357**
   **2, 34554**

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Write a C program to **find the power shortage of the 30th day**.

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